

Artificial Intelligence & Machine Learning

Task 3: Model Validation, Overfitting Control & Hyperparameter Tuning

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1 Objective

The objective of this task was to improve machine learning model reliability through model validation, overfitting detection, cross-validation, and hyperparameter tuning. The California Housing Dataset was used to develop and evaluate regression models for house price prediction.

2 Dataset Description

The California Housing Dataset from Scikit-learn was used.

2.1 Input Features

- MedInc
- HouseAge
- AveRooms
- AveBedrms
- Population
- AveOccup
- Latitude
- Longitude

2.2 Target Variable

- Median House Value (HousePrice)

3 Data Preprocessing

The following preprocessing steps were performed:

1. Loaded the California Housing Dataset.
2. Separated input features and target variable.
3. Applied StandardScaler for feature normalization.
4. Split the dataset into training and testing sets using an 80:20 ratio.

4 Baseline Models

Two baseline models from Task 2 were used for comparison.

4.1 Linear Regression

Linear Regression establishes a linear relationship between independent variables and the target variable.

Linear Regression Results

Metric	Value
RMSE	0.745581
R ² Score	0.575788

4.2 Ridge Regression

Ridge Regression introduces L2 regularization to reduce model variance.

Ridge Regression Results

Metric	Value
RMSE	0.745554
R ² Score	0.575819

5 Overfitting Analysis

A Decision Tree Regressor was trained without constraints.

5.1 Training Performance

$$\text{Train RMSE} = 3.218325866275131 \times 10^{-16}$$

5.2 Testing Performance

$$\text{Test RMSE} = 0.7030445773467542$$

5.3 Analysis

The training RMSE is almost zero, indicating that the model perfectly memorized the training data. However, the testing RMSE is significantly larger.

The large gap between training and testing performance clearly indicates overfitting.

6 Cross Validation

6.1 Purpose

Cross-validation provides a more reliable estimate of model performance than a single train-test split.

A 5-Fold Cross Validation strategy was used.

6.2 Cross Validation Scores

```
[-0.78796899  
-0.68527964  
-0.80721441  
-0.89315807  
-0.84548406]
```

6.3 Discussion

The scores remain relatively consistent across all folds. This indicates that the model behaves similarly across different subsets of the dataset and demonstrates stable performance.

7 Hyperparameter Tuning

To reduce overfitting and improve model generalization, GridSearchCV was used.

7.1 Parameter Grid

```
{  
'max_depth': [3, 5, 7, 10],  
'min_samples_split': [2, 5, 10]  
}
```

7.2 Best Parameters Found

```
{  
'max_depth': 10,  
'min_samples_split': 10  
}
```

7.3 Interpretation

- **max_depth = 10** prevents excessive tree growth.

- `min_samples_split = 10` reduces unnecessary splits and helps prevent overfitting.

8 Evaluation of Tuned Decision Tree

After optimization, the tuned model was evaluated on the test dataset.

Tuned Decision Tree Results

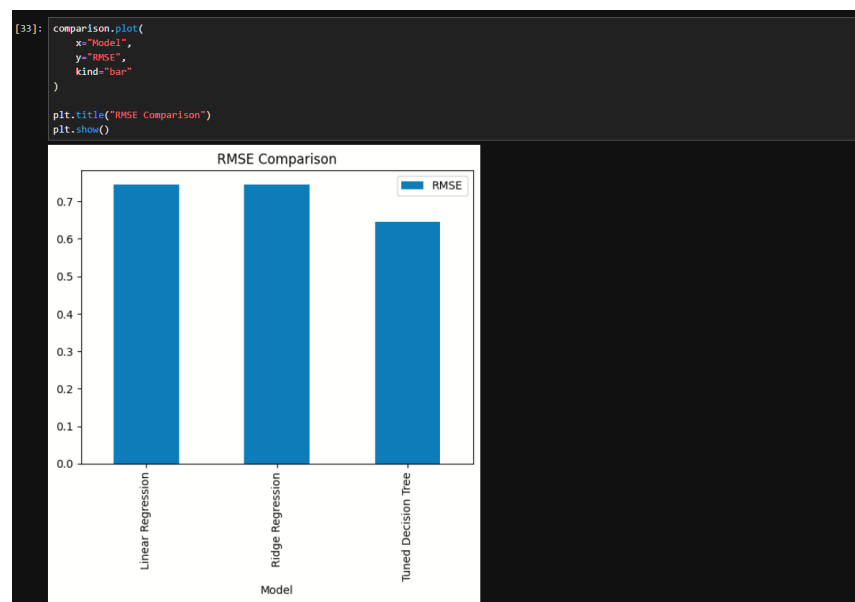
Metric	Value
RMSE	0.645430
R ² Score	0.682099

9 Model Comparison

Comparison of All Models

Model	RMSE	R ² Score
Linear Regression	0.745581	0.575788
Ridge Regression	0.745554	0.575819
Tuned Decision Tree	0.645430	0.682099

10 RMSE Comparison Graph



RMSE Comparison Among Models

11 Final Model Selection

The Tuned Decision Tree was selected as the final model because:

- It achieved the lowest RMSE (0.645430).
- It achieved the highest R^2 score (0.682099).
- Hyperparameter tuning successfully reduced overfitting.
- Cross-validation confirmed stable performance.
- It generalized better than the untuned model.

12 Conclusion

This task demonstrated the importance of model validation and optimization in machine learning.

The initial Decision Tree model exhibited severe overfitting. Cross-validation provided reliable performance estimation, while GridSearchCV optimized model parameters and improved generalization.

Among all evaluated models, the Tuned Decision Tree achieved the best balance between predictive accuracy and generalization performance, making it the most suitable model for house price prediction.

13 References

1. Scikit-learn Documentation
2. California Housing Dataset Documentation
3. GridSearchCV Documentation
4. Cross Validation Documentation
5. Machine Learning Model Evaluation Techniques